

### Practice Sheet: Mathematical Expectation

1. Given the following table:

|        |      |      |      |   |      |      |      |
|--------|------|------|------|---|------|------|------|
| $x$    | -3   | -2   | -1   | 0 | 1    | 2    | 3    |
| $p(x)$ | 0.05 | 0.10 | 0.30 | 0 | 0.30 | 0.15 | 0.10 |

Compute: (i)  $E(X)$ , (ii)  $E(2X \pm 3)$ , (iii)  $E(4X + 5)$ , (iv)  $E(X^2)$ .

2.  $A$  and  $B$  throw with one dice for a stake of ₹ 44,000 which is to be won by the player who first throws a 6. If  $A$  has the first throw, what are their respective expectations ?
3. Let  $X$  be a random variable with the following probability distribution:

|            |               |               |               |
|------------|---------------|---------------|---------------|
| $x$        | -3            | 6             | 9             |
| $P(X = x)$ | $\frac{1}{6}$ | $\frac{1}{2}$ | $\frac{1}{3}$ |

Find  $E(X)$  and  $E(X^2)$  and using the laws of expectation, evaluate  $E(2X + 1)^2$ .

4. (a) Find the expectation of the number on a dice when thrown.  
(b) Two unbiased dice are thrown. Find the expected values of the sum of numbers of points on them.
5. In four tosses of a coin, let  $X$  be the number of heads. Tabulate the 16 possible outcomes with the corresponding values of  $X$ . By simple counting, derive the probability distribution of  $X$  and hence calculate the expected value of  $X$ .
6. A box contains 3 red, 4 blue, and 5 green balls. One ball is drawn at random. Let  $X$  be defined as:  $X = \begin{cases} 1, & \text{if a red ball is drawn} \\ 2, & \text{if a blue ball is drawn} \\ 3, & \text{if a green ball is drawn} \end{cases}$

Find  $E(X)$ .

7. A fair coin is tossed three times. Let  $X$  denote the number of tails obtained. Find  $E(X)$ .
8. A discrete random variable  $X$  has probability mass function:  $P(X = k) = \frac{1}{10}$ ,  $k = 1, 2, 3, \dots, 10$  Find  $E(X)$  and  $E(X^2)$ .
9. A random variable  $X$  represents the score obtained when drawing one card from a deck, where Ace = 1, 2–10 as numbers, Jack = 11, Queen = 12, King = 13 (all equally likely). Find  $E(X)$ .
10. A game is played by tossing a fair die. If a 6 appears, you win \$10; otherwise, you lose \$2. Find the expected value of your winnings.
11. The probability density function (pdf) of  $X$  is  $f(x) = \frac{1}{2}$ ,  $0 \leq x \leq 2$ . Find  $E(X)$ .
12. A continuous random variable  $X$  has pdf  $f(x) = 2x$ ,  $0 \leq x \leq 1$ . Find  $E(X)$  and  $E(X^2)$ .
13. A random variable  $X$  has pdf  $f(x) = \frac{3}{8}x^2$ ,  $0 \leq x \leq 2$ . Find  $E(X)$ .